



SYDNEY BOYS HIGH SCHOOL
MOORE PARK, SURRY HILLS

2003

**HIGHER SCHOOL CERTIFICATE
ASSESSMENT TASK # 2**

Mathematics Extension 1

General Instructions

- Reading time – 5 minutes.
- Working time – 90 minutes.
- Write using black or blue pen.
- Board approved calculators may be used.
- All necessary working should be shown in every question if full marks are to be awarded.
- Marks may **NOT** be awarded for messy or badly arranged work.
- Hand in your answer booklets in 3 sections. Section A (Questions 1 and 2), Section B (Questions 3 and 4) and Section C (Questions 5 and 6).
- Start each **NEW** section in a separate answer booklet.

Total Marks - 90 Marks

- Attempt Sections A - C
- All questions are NOT of equal value.

Examiner: *C. Kourtesis*

This is an assessment task only and does not necessarily reflect the content or format of the Higher School Certificate.

SECTION A

Question 1 (15 marks)		Marks
(a)	Find $\sin 1 \cdot 4$	1
(b)	Express 450° in radians (in terms of π).	1
(c)	Differentiate	
(i)	$\sin 4x$	1
(ii)	$(3x + 1)^{20}$	1
(iii)	$\cos(1 - 2x)$	1
(d)	Find the gradient of the tangent to the curve $y = 4x^{3/2}$ at the point $(1, 5)$.	2
(e)	If $y = x \tan x$, use the product rule to find $\frac{dy}{dx}$.	2
(f)	If $f(t) = \frac{1}{t}$, find the value of $f''(4)$.	2
(g)	Find	
(i)	$\int_1^9 (1 + \sqrt{x}) dx$	2
(ii)	$\int_0^{2\pi} \cos 2x dx$	2

SECTION A (continued)

Question 2 (16 marks)			Marks
(a)	(i)	Sketch the graph of the function $y = (x + 1)^2$.	1
	(ii)	State the values of x for which the function is decreasing.	1
(b)	(i)	Sketch the graph of $y = \sin x$ for $0 \leq x \leq 2\pi$.	1
	(ii)	Find the area bounded by the curve $y = \sin x$ and the x axis for $0 \leq x \leq 2\pi$.	2
(c)		Find a primitive of $\frac{x + \sqrt{x}}{x}$.	2
(d)		In how many ways can:	
	(i)	6 different books be arranged on a shelf?	1
	(ii)	3 different paintings be hung in a row, from a collection of 8?	2
(e)		Find $\lim_{x \rightarrow 0} \frac{\sin 7x}{5x}$	2
(f)		The curve $y = 5x^4 - bx^2$ has a turning point at $x = 1$.	2
		Find the value of b .	
(g)		If $g(x) = \sin^3 x$, find $g'\left(\frac{\pi}{4}\right)$	2

SECTION B (Start a NEW booklet)

Question 3 (14 marks)

Marks

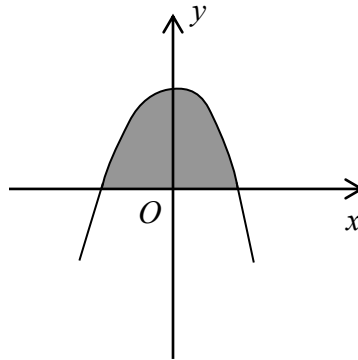
- (a) For a certain curve $\frac{dy}{dx} = 2x - 5$ and $(2, -18)$ lies on the curve.

3

Find the equation of the curve.

- (b)

3



The shaded region lying between the curve $y = 4 - x^2$ and the x - axis is rotated about the x - axis.

Find the volume of the resulting solid.

- (c) Consider the curve $y = x^4 + 4x^3 - 16x + 1$

- (i) Show that it has a minimum turning point at $x = 1$.

3

- (ii) Find the values of x for which the curve is concave up.

2

- (d) Prove by mathematical induction that, for all positive integers n , $3^{2^n} - 1$ is divisible by 8.

3

SECTION B (continued)

Question 4 (15 marks)

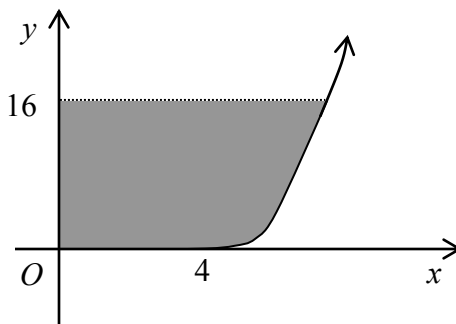
Marks

- (a) A committee of 3 men and 2 women is to be selected from a group of 10 men and 8 women.

2

In how many ways can this be done?

- (b)



The shaded area bounded by the curve $y = (x-4)^3$, the coordinate axes and the lines $y = 0$ and $y = 16$, is rotated about the y -axis.

- (i) Show that the volume V of the solid formed is given by

2

$$V = \pi \int_0^{16} (y^{2/3} + 8y^{1/3} + 16) dy$$

- (ii) Find the volume V in terms of π

3

- (c) Consider the function $f(x) = \frac{x}{x+1}$

- (i) Find $f'(x)$.

2

- (ii) Show that there are NO stationary points.

1

- (iii) Show that the function is always increasing.

1

- (iv) Write down the equations of any asymptotes.

2

- (v) Sketch the graph of $y = f(x)$.

2

SECTION C (Start a NEW booklet)

Question 5 (15 marks)

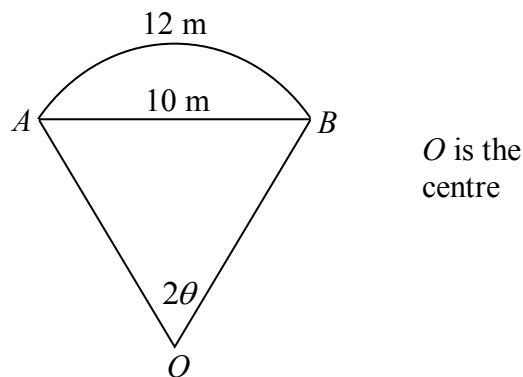
Marks

(a) (i) Show that $\cos 2x = 1 - 2\sin^2 x$ 2

(ii) Hence find $\int \sin^2 4x \, dx$ 3

(b) Find the derivative of $\sin(\sqrt{\tan 2x})$ 2

(c)



A pipe which is 12 m long is bent into a circular arc which subtends an angle of 2θ at the centre of the circle.
The chord of the circle joining the ends of the arc is 10 m long.

(i) Show that $6\sin\theta - 5\theta = 0$ 4

(ii) Show that $\theta_0 = 1^\circ$ is a good first approximation to the value of θ . 1

(iii) Use one application of Newton's Method to find another approximation to θ . 3

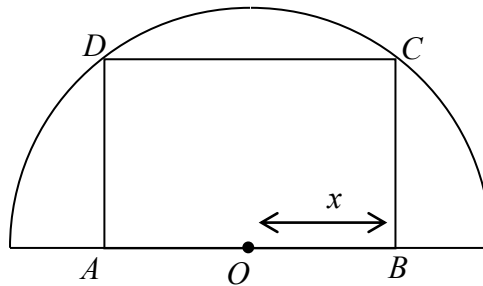
SECTION C (continued)

Question 6 (15 marks)

Marks

- (a) Prove by mathematical induction that $4^n \geq n^2$ for all positive integers n 4

(b)



A rectangle $ABCD$ is inscribed in a semi-circle of radius 10 units, centre O . Let $OB = x$.

- (i) Show that the area A square units of the rectangle $ABCD$ is given by 2

$$A = 2x\sqrt{100 - x^2}$$

- (ii) Find the values of x for which $\frac{dA}{dx} = 0$ 3

- (iii) The rectangle $ABCD$ has a maximum area. 2

Find the maximum area.

[You are **NOT** required to prove that the area is a maximum]

- (c) Given the curve $y = x^{1/3} + \frac{1}{4}x^{4/3}$ find, giving reasons, any points of inflexion. 4

THIS IS THE END OF THE PAPER

STANDARD INTEGRALS

$$\int x^n dx = \frac{1}{n+1} x^{n+1}, n \neq -1; x \neq 0, \text{ if } n < 0$$

$$\int \frac{1}{x} dx = \ln x, x > 0$$

$$\int e^{ax} dx = \frac{1}{a} e^{ax}, a \neq 0$$

$$\int \cos ax dx = \frac{1}{a} \sin ax, a \neq 0$$

$$\int \sin ax dx = -\frac{1}{a} \cos ax, a \neq 0$$

$$\int \sec^2 ax dx = \frac{1}{a} \tan ax,$$

$$\int \sec ax \tan ax dx = \frac{1}{a} \sec ax, a \neq 0$$

$$\int \frac{1}{a^2 + x^2} dx = \frac{1}{a} \tan^{-1} \frac{x}{a}, a \neq 0$$

$$\int \frac{1}{\sqrt{a^2 - x^2}} dx = \sin^{-1} \frac{x}{a}, a > 0, -a < x < a$$

$$\int \frac{1}{\sqrt{x^2 - a^2}} dx = \ln \left(x + \sqrt{x^2 - a^2} \right) x > a > 0$$

$$\int \frac{1}{\sqrt{x^2 + a^2}} dx = \ln \left(x + \sqrt{x^2 + a^2} \right)$$

NOTE: $\ln x = \log_e x, x > 0$

May/June 2003 3 unit

Question 1

(a) $\sin 1.4 = 0.9854$ 1

(b) $450^\circ = \frac{450 \times \pi}{180}$
 $= \frac{5\pi}{2}$ 1

(c) (i) $\frac{d}{dx} \sin 4x = 4 \cos 4x$ 1

(ii) $\frac{d}{dx} (3x+1)^{20} = 20(3x+1)^{19} \times 3$
 $= 60(3x+1)^{19}$ 1

(iii) $\frac{d}{dx} \cos(1-2x) = -\sin(1-2x)(-2)$
 $= 2 \sin(1-2x)$ 1

(d) $y = 4x^{\frac{3}{2}}$ (1.5)

$y' = \frac{3}{2} \times 4x^{\frac{1}{2}}$
 $= 6x^{\frac{1}{2}}$

$y'(1) = 6$

$\therefore \text{Gradient} = 6$ 2

(e) $y = x \tan x$

$\frac{dy}{dx} = \tan x + x \sec^2 x$ 2

$f(t) = \frac{1}{t}$

$= t^{-1}$

$f'(t) = -t^{-2}$

$f''(t) = 2t^{-3}$

$= \frac{2}{t^3}$

$f''(4) = \frac{1}{32}$ 2

(g) (i) $\int_1^9 (1+\sqrt{x}) dx$

$= \left[x + \frac{2x^{\frac{3}{2}}}{3} \right]_1^9$

$= (9+18) - (1+\frac{2}{3})$

$= 25\frac{1}{3}$

$= \frac{76}{3}$ 2

(ii) $\int_0^{2\pi} \cos 2x dx$

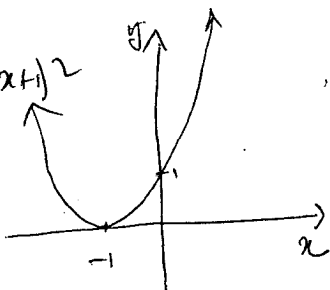
$= \frac{1}{2} [\sin 2x]_0^{2\pi}$

$= \frac{1}{2} [0 - 0]$

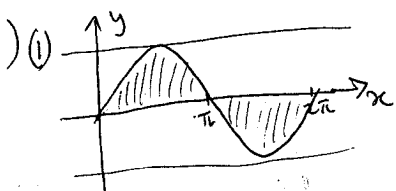
$= 0$ 2

Question 2

1) $y = (x+1)^2$



Decreasing = $\{x: x \leq -1\}$



$$\begin{aligned} \text{Area} &= 2 \int_0^{\pi} \sin x \, dx \\ &= 2 [-\cos x]_0^{\pi} \\ &= 2 [-(-1) - (-1)] \\ &= 4 \text{ sq. units} \end{aligned}$$

$$\begin{aligned} \int \frac{x + \sqrt{x}}{x} dx &= \int (1 + x^{-1/2}) dx \\ &= x + \frac{x^{1/2}}{1/2} + C \\ &= x + 2\sqrt{x} + C \end{aligned}$$

(a) (i) $6! = 720$

(ii) ${}^8C_3 \times 3! = 56 \times 6 = 336$

(c) $\lim_{x \rightarrow 0} \frac{\sin 7x}{5x} = \frac{7}{5} \lim_{x \rightarrow 0} \frac{\sin 7x}{7x} = \frac{7}{5} \times 1 = \frac{7}{5}$

(d) $y = 5x^4 - bx^2$
 $y' = 20x^3 - 2bx$
 T.P. at $x = 1 \therefore y'(1) = 0$
 $\therefore 0 = 20(1)^3 - 2b(1)$
 $0 = 20 - 2b$

$2b = 20$
 $b = 10$

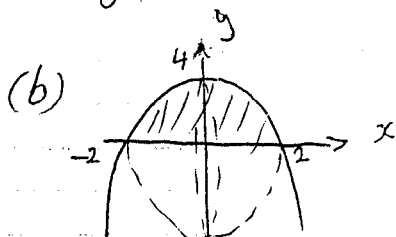
(g) $g(x) = \sin^3 x$
 $g'(x) = 3 \sin^2 x \cos x$
 $g'(\frac{\pi}{4}) = 3 \sin^2 \frac{\pi}{4} \cos \frac{\pi}{4}$
 $= 3 (\frac{1}{\sqrt{2}})^2 (\frac{1}{\sqrt{2}})$
 $= \frac{3}{2\sqrt{2}}$

Solutions Maths ext 1 4R12 2003 Task #2 May

(3) (a) $y' = 2x - 5$
 $y = \int (2x - 5) dx$
 $y = x^2 - 5x + C$

(2-18) $-18 = 4 - 10 + C$
 $-18 = -6 + C$
 $-18 + 6 = C$
 $C = -12$

$\therefore y = x^2 - 5x - 12 //$



$V = \pi \int_{-2}^2 y^2 dy$
 $= \pi \int_{-2}^2 (4 - x^2)^2 dy$

(c) $y = x^4 + 4x^3 - 16x + 1$

(i) $y' = 4x^3 + 12x^2 - 16$
 $y'' = 12x^2 + 24x$

When $y' = 0$ $4x^3 + 12x^2 - 16 = 0$

if $x = 1$, $4 + 12 - 16 = 0$

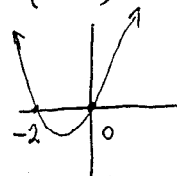
at $x = 1$, $y'' = 36 > 0$

at $x = 1$, min. pt. (3)

(3)

$y = 4 - x^2$
 So $x^2 = 4 - y$

(ii) when $y'' > 0$
 $12x^2 + 24x > 0$
 $12x(x + 2) > 0$



When $x < -2$, $x > 0$ (2)

$= \pi \int_{-2}^2 (16 - 8x^2 + x^4) dx$
 $= \pi \left[16x - \frac{8x^3}{3} + \frac{x^5}{5} \right]_{-2}^2$

$= \pi \left[32 - \frac{64}{3} + \frac{32}{5} - \left(-32 + \frac{64}{3} - \frac{32}{5} \right) \right]$

(3) $= \pi \left[34 \frac{2}{15} \right]$ (3)
 $512 \pi / 15^3$

3 (d) step 1 let $n=1$, $8 \mid (3^2 - 1)$ is true.
when $n=1$, step 1 is true.

step 2 If there exists a value $n=k$, where k is a positive integer then assume that $8 \mid (3^{2k} - 1)$ and we must prove that for $n=k+1$, $8 \mid (3^{2(k+1)} - 1)$ is true.

$$\begin{aligned}\text{Now } 3^{2k+2} - 1 &= 3^{2k} \cdot 3^2 - 1 \\ &= 3^{2k} \times 9 - 1 \\ &= 9(3^{2k} - 1) + 8\end{aligned}$$

$$\begin{aligned}\text{now } 8 \mid 9(3^{2k} - 1) \text{ and } 8 \mid 8 \\ 8 \mid (3^{2(k+1)} - 1)\end{aligned}$$

So $n=k+1$ is true.

step 3 We have assumed it true for $n=1$, and $n=k$ and proved it true for $n=k+1$.
Hence statement is true for $n=2, n=3$ etc.
 $\forall$ +ve integers n .

(3)

B ④ (a) ${}^{10}C_3 \times {}^8C_2$ $120 \times 28 = 3360$ ②

(b) (i) $V = \pi \int_0^{16} x^2 dy$ $y = (x-4)^3$
 $= \pi \int_0^{16} (y^{\frac{1}{3}} + 4)^2 dy$ $y^{\frac{1}{3}} = x-4$
 $x = y^{\frac{1}{3}} + 4$
 $= \pi \int_0^{16} (y^{\frac{2}{3}} + 8y^{\frac{1}{3}} + 16) dy$ ②

(ii) $V = \pi \left[\frac{3}{5} y^{\frac{5}{3}} + 6y^{\frac{4}{3}} + 16y \right]_0^{16}$
 $V = \pi \left[\frac{3}{5} \times 16^{\frac{5}{3}} + 6 \times 16^{\frac{4}{3}} + 16 \times 16 \right]$
 $= \pi \left[\frac{3}{5} \cdot (\sqrt[3]{16})^5 + 6 \times (\sqrt[3]{16})^4 + 256 \right]$
 $= \pi [60.956 + 241.905 + 256]$
 $\doteq 558.861 \pi u^3$ ③

(c) $f(x) = \frac{x}{x+1}$

(i) $f'(x) = \frac{(x+1) \times 1 - x \times 1}{(x+1)^2} = \frac{1}{(x+1)^2}$ ②

(ii) $\frac{1}{(x+1)^2} = \frac{0}{1}$ x has no solution ①

(iii) except for $x = -1$, $y' = \frac{1}{(x+1)^2} > 0$ always ①

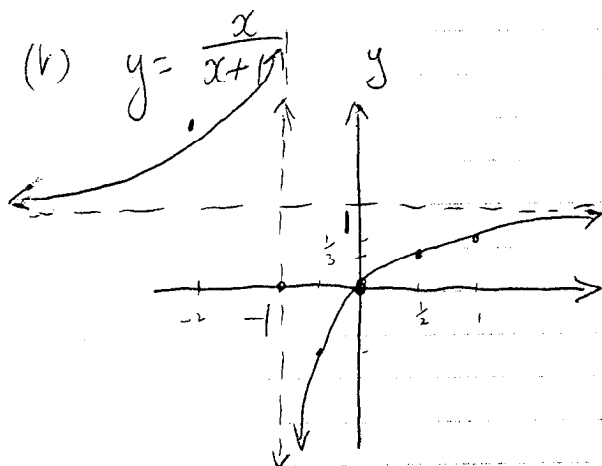
(iv) $f(x) = \frac{x}{x+1}$

$x = -1$ is an asymptote

$$f(x) = \frac{x}{x(1+\frac{1}{x})} = \frac{1}{1+\frac{1}{x}}$$

as $x \rightarrow \pm\infty$, $\frac{1}{x} \rightarrow 0$ $f(x) = 1$ is another asymptote

(2)



(2)

15 marks.

Section C — Solutions

Question (5)

(a) (i) $\cos 2x = \cos(x+x)$
 $= \cos x \cos x - \sin x \sin x$

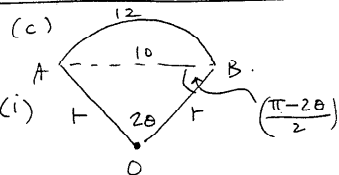
[2] $= 1 - \sin^2 x - \sin^2 x$
 $= 1 - 2\sin^2 x$

$\therefore \sin^2 x = \frac{1}{2}(1 - \cos 2x)$

(ii) $\int \sin^2 x \, dx$
 $= \frac{1}{2} \int (1 - \cos 2x) \, dx$
 $= \frac{x}{2} - \frac{\sin 2x}{2} + C$

(b) $\frac{d}{dx} \sin(\sqrt{\tan x})$
 $= \cos \sqrt{\tan x} \cdot \frac{1}{2} (\tan x)^{-\frac{1}{2}}$

[2] $\cdot \frac{1}{2} \sec^2 x$
 $= \cos \sqrt{\tan x} \cdot \sec^2 x$
 $= \frac{\cos \sqrt{\tan x} \cdot \sec^2 x}{\sqrt{\tan x}}$



Let r be the radius. $[l = r\theta]$
 $12 = r(2\theta)$
 $\therefore r = \frac{6}{\theta}$ — (1)

In $\triangle AOB$, apply sine rule

$\frac{10}{\sin 2\theta} = \frac{r}{\sin(\frac{\pi}{2} - \theta)}$

$\frac{10}{2\sin\theta\cos\theta} = \frac{r}{\cos\theta}$

$\therefore r = \frac{5}{\sin\theta}$ — (2)

Equate (1) and (2)

$\frac{6}{\theta} = \frac{5}{\sin\theta}$

$\therefore 6\sin\theta = 5\theta$

i.e. $6\sin\theta - 5\theta = 0$

(ii) When $\theta = 1$

$f(\theta) = 6\sin(\theta) - 5\theta$

[1] $f(1) = 0.0488$

$f'(1) = -1.7582$

$[f'(0) = 6\cos 0 - 5]$

[3] $\theta_1 = 0 - \frac{f(0)}{f'(0)}$
i.e. $\theta_1 = 1 - \frac{f(1)}{f'(1)}$

$\therefore \theta_1 = 1 + \frac{0.0488}{1.758}$
 $= 1.0278$

Question (6)

Let $S(n)$ be the statement that $4^n \geq n^2$

For $n=1$, $4 > 1^2$
 $\therefore S(1)$ is true.

Assume $S(k)$ is true

[4] i.e. $4k \geq k^2$

Consider $n=k+1$

$4^{k+1} = 4 \cdot 4^k$

$> 4k^2$

$= k^2 + 3k^2$

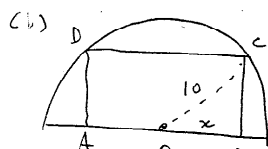
Now, $3k^2 \geq 2k+1$

Whenever $k \geq 1$

$\therefore 4^{k+1} \geq k^2 + 2k + 1$

i.e. $4^{k+1} \geq (k+1)^2$

Since $S(n)$ is true for $n=1$ and true for $n=k+1$ when true for $n=k$ ($k \in \mathbb{Z}^+$) $\therefore$ statement is true



(i) $t=10$, In $\triangle OBC$

[2] $10^2 = x^2 + BC^2$
 $BC = \sqrt{100 - x^2}$

Area of $ABCD = 2x\sqrt{100 - x^2}$

(ii) $\frac{dA}{dx} = 2\sqrt{100 - x^2} + 2x \cdot \frac{1}{2} (100 - x^2)^{-\frac{1}{2}} \cdot (-2x)$

$\therefore \frac{dA}{dx} = 2\sqrt{100 - x^2} - \frac{2x^2}{\sqrt{100 - x^2}}$

[3] $\frac{dA}{dx} = 0$ $100 - x^2 = x^2$

$\therefore 2x^2 - 100 = 0$

$\therefore x^2 = 50$ $x = 5\sqrt{2}$

$\therefore A_{\max} = 2 \times 5\sqrt{2} \times 5\sqrt{2}$

[2] $= 10\sqrt{2}(\sqrt{50})$

$= 100$

(c) $y = x^{\frac{1}{3}}(1 + \frac{1}{4}x)$

$\frac{dy}{dx} = \frac{1}{3}x^{-\frac{2}{3}}(1 + \frac{x}{4}) + \frac{1}{4}x^{\frac{1}{3}}$

[1] $= \frac{1}{3}(x^{-\frac{2}{3}} + x^{\frac{1}{3}})$

$\frac{d^2y}{dx^2} = \frac{1}{3}[-\frac{2}{3}x^{-\frac{5}{3}} + \frac{1}{3}x^{-\frac{2}{3}}]$

$= \frac{x^{-\frac{5}{3}}}{9}(x - 2)$

$\frac{d^2y}{dx^2} = 0 \Rightarrow x = 2$

$\frac{d^2y}{dx^2} = 0 \Rightarrow x = 2$

x	1	2	3	-1
$f''(x)$	< 0	0	> 0	> 0

When $x = 2$

$y = \sqrt[3]{2}(1 + \frac{1}{2})$

$= \frac{3}{2}\sqrt[3]{2}$

$= 3 \cdot 2^{-\frac{2}{3}}$

$\therefore (2, 3 \cdot 2^{-\frac{2}{3}})$ is a point of